Ronald A. Cardenas Acosta

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Professional Summary

Applied Scientist with both deep and broad knowledge in generative technologies with over 6 years of experience in research and over 2 years delivering LLM-based and agentic solutions (text, image, speech) to a diverse set of end-users, with publications in top-tier conferences such as ACL, EMNLP, NAACL, ICLR.

Skills

Programming Languages: Python, C/C++, Java, R, Perl, Bash, JavaScript, Ruby
Deep Learning Frameworks: PyTorch, TensorFlow, JAX, TRL, Transformers
Distributed training, inference: Ray, vLLM, SGLang, veRL

Professional Experience

Research Scientist London, UK
Huawei Research Centre 05/2024 – To date

- Implemented and trained deep research agents capable of information seeking, web navigation, and coding, aggregating multi-modal data (text, image, video, speech). Our agent reached top-1 in the GAIA leaderboard.
- Collaborated in the design and curation of a new benchmark for deep research agents that focuses on the evaluation of intermediate results, in which all current agents struggle.
- Delivered a reasoning strategy that reduces token usage by 22% without accuracy degradation, by training the model to switch to and from latent space reasoning and token space reasoning.
- Proposed memory management strategies for coding agents, improving performance up to 5% in SWE Bench
- Developed a new preference optimization technique that controls the reward contribution of response tokens via sparse token masks, improving up to 16% on reasoning tasks, 5% in summary quality, and 25% in helpfulness & harmlessness control.
- Developed an inference pipeline that improves autoformalization of mathematical problems into Lean4 by leveraging test-time reasoning strategies, in collaboration with mathematicians, resulting in 10% more formalized problems.

Applied Science Intern Cambridge, UK
Amazon Alexa 06/2019 - 08/2019
Investigated transfer learning strategies across multi-modal tasks involving question domain categorization, speaker identification, and image retrieval.

Research Associate Lima, Peru
Telecommunication Technologies Lab 02/2016 - 06/2017
Developed and deployed a speech recognition system for Quechua language, which involved collecting and curating data, training the model and deploying at scale.

Education

Centre for Doctoral Training in NLP, University of Edinburgh Edinburgh, UK
Ph.D. in Computer Science 03/2024
Thesis: Cognitive Structures of Content for Controlled Summarization

Erasmus Joint Programme Czech Republic & Malta
Charles University, Prague & University of Malta
M.Sc. in Language and Communication Technologies 06/2019
Thesis: Universal Morphological Analysis using Reinforcement Learning

National University of Engineering Lima, Peru
B.Sc. in Mechatronics Engineering 12/2015
Summa Cum Laude

Relevant Research Experience

Institute for Language, Cognition, and Communication, University of Edinburgh 09/2019-03/2024

- Investigated methods to give human users straightforward control over text properties of generated summaries, in scenarios where text inputs are extremely long and highly redundant.

NAVER Labs, Grenoble, France
10/2022-03/2023

- Worked on controlled generation in large language models using distributional policy gradients, in the case of factuality and coherence control.

IBM, Dublin, Ireland
09/2021-04/2022

- Developed a framework to assist journalists in writing press release summaries of scientific articles, and curated a novel dataset for associated NLP tasks.

Leadership & Activities

HackSpace Lima, Peru
Lecturer 07 – 11/2016

- Organised and delivered a 32-hour mini-course on introductory topics to Machine Learning and NLP aimed at IT professionals.

Artificial Intelligence Student Association, National University of Engineering **Lima**, **Peru**
President 02/2014 – 12/2015

- Organising seminars and workshops to introduce undergraduates to research projects in Machine Learning.
- Coached teams for competitive programming, participating in contests such as ACM ICPC, IEEExtreme

Additional Training

- Mediterranean Machine Learning Summer School, M2L. 2022
- Lisbon Machine Learning Summer School, LxMLS,. 2014, ‘20, ‘21, ‘22
- Advanced Language Processing Winter School, ALPS. 2021, ‘22, ‘23
- Machine Learning Summer School. 2015
- Competitive Programming Summer School, University of Campinas, Brazil. 2013

References

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